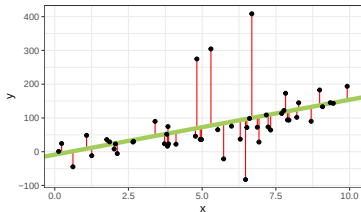
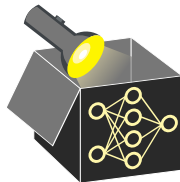


Interpretable Machine Learning

Interpretable Models 1

Linear Regression Model



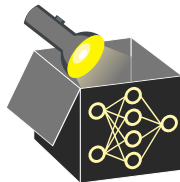
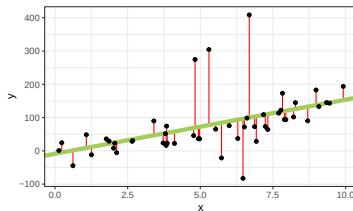
Learning goals

- LM basics and assumptions
- Interpretation of main effects in LM
- What are significant features?

LINEAR REGRESSION

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_p x_p + \epsilon = \mathbf{x}^\top \boldsymbol{\theta} + \epsilon$$

- y : target / output
- ϵ : remaining error / residual
- θ_j : weight of input feature x_j (intercept θ_0)
 $\leadsto$ model consists of $p + 1$ weights



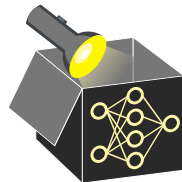
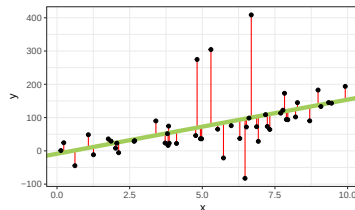
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Properties and assumptions ▶ "Faraway, Ch. 7" 2002

- **Linear** relationship between features and target



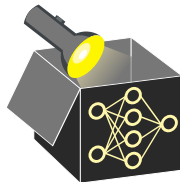
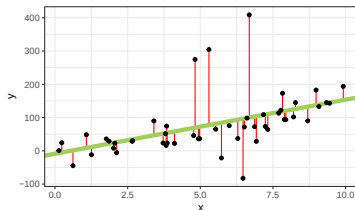
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Properties and assumptions ► "Faraway, Ch. 7" 2002

- **Linear** relationship between features and target
- ϵ and $y|\mathbf{x}$ are **normally** distributed with **constant variance** (homoscedastic)
 $\rightsquigarrow \epsilon \sim N(0, \sigma^2) \Rightarrow (y|\mathbf{x}) \sim N(\mathbf{x}^\top \boldsymbol{\theta}, \sigma^2)$
 $\rightsquigarrow$ if violated, inference-based metrics (e.g., p-values) are invalid



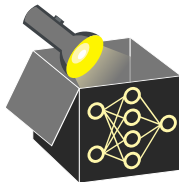
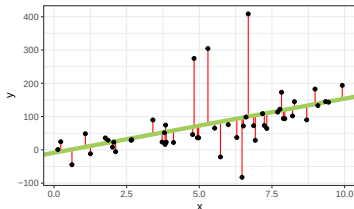
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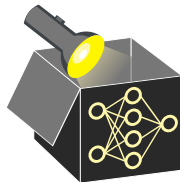
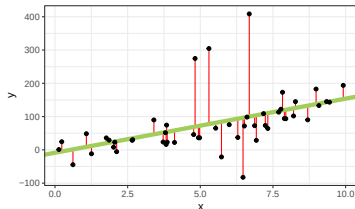
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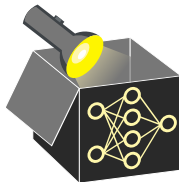
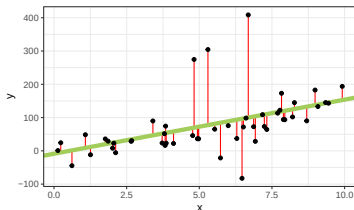
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- Independence of observations (e.g., no repeated measurements)
- Features x_j independent from error term ϵ
- No or little multicollinearity (i.e., no strong feature correlations)

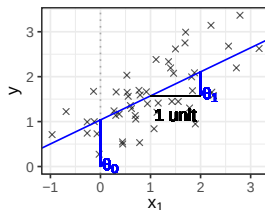
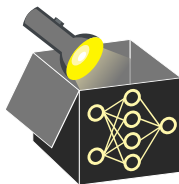


LINEAR REGRESSION - INTERPRETATION

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_p x_p + \epsilon = \mathbf{x}^\top \boldsymbol{\theta} + \epsilon$$

Interpretation of weights (**feature effects**) depend on type of feature:

- **Numerical** x_j : Increasing x_j by one unit changes outcome by θ_j , *ceteris paribus* (*ceteris paribus* (c.p.) means "everything else held constant".)

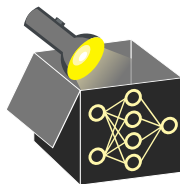
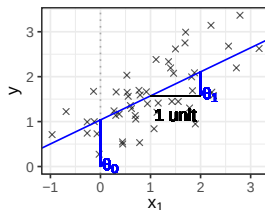


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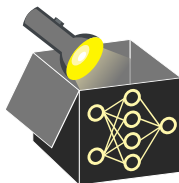
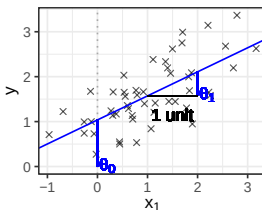


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- **Categorical feature x_j with L categories:**
 - Create $L - 1$ one-hot-encoded features $x_{j,1}, \dots, x_{j,L-1}$ (each having its own weight)
 - Left out cat. is reference ($\hat{=}$ dummy encoding)
 - $\rightsquigarrow$ Interpretation: Outcome changes by $\theta_{j,i}$ for category i compared to reference cat., c.p.

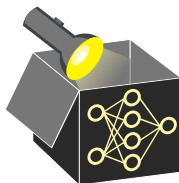
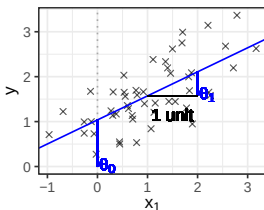


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- **Intercept** θ_0 : Expected outcome if all feature values are set to 0



LINEAR REGRESSION - INTERPRETATION

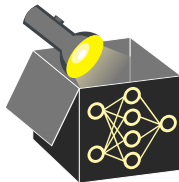
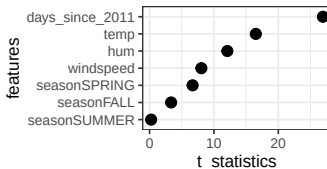
$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_p x_p + \epsilon = \mathbf{x}^\top \boldsymbol{\theta} + \epsilon$$

Feature importance:

- Absolute **t-statistic** value: $\hat{\theta}_j$ scaled with standard error ($SE(\hat{\theta}_j) \triangleq$ reliability of estimate)

$$|t_{\hat{\theta}_j}| = \left| \frac{\hat{\theta}_j}{SE(\hat{\theta}_j)} \right|$$

- High t -values $\Rightarrow$ important (significant) feat.



LINEAR REGRESSION - INTERPRETATION

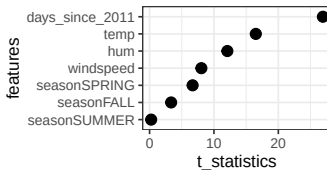
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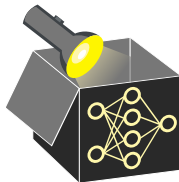
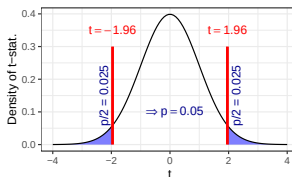
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- High t -values $\Rightarrow$ important (significant) feat.
- **p-value**: probability of obtaining a more extreme test statistic assuming H_0 is correct (here: $\theta_j = 0$, i.e., feat. j not significant)
 $\leadsto$ High $|t| \Rightarrow$ small p-val. (speak against H_0)



$H_0: \theta_j = 0$ vs. $H_1: \theta_j \neq 0$ (two-sided)

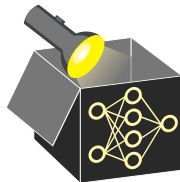


EXAMPLE: LIN. REGRESSION - MAIN EFFECTS

Bike data: predict no. of rented bikes using 4 numeric, 1 cat. feat. (season)

$$\begin{aligned}\hat{y} = & \hat{\theta}_0 + \hat{\theta}_1 \mathbb{1}_{x_{\text{season}}=\text{SPRING}} + \\ & \hat{\theta}_2 \mathbb{1}_{x_{\text{season}}=\text{SUMMER}} + \\ & \hat{\theta}_3 \mathbb{1}_{x_{\text{season}}=\text{FALL}} + \hat{\theta}_4 x_{\text{temp}} + \\ & \hat{\theta}_5 x_{\text{hum}} + \hat{\theta}_6 x_{\text{windspeed}} + \\ & \hat{\theta}_7 x_{\text{days_since_2011}}\end{aligned}$$

	Weights	SE	t-stat.	p-val.
(Intercept)	3229.3	220.6	14.6	0.00
seasonSPRING	862.0	129.0	6.7	0.00
seasonSUMMER	41.6	170.2	0.2	0.81
seasonFALL	390.1	116.6	3.3	0.00
temp	120.5	7.3	16.5	0.00
hum	-31.1	2.6	-12.1	0.00
windspeed	-56.9	7.1	-8.0	0.00
days_since_2011	4.9	0.2	26.9	0.00

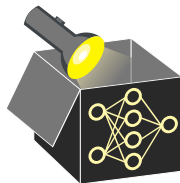


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Interpretation:

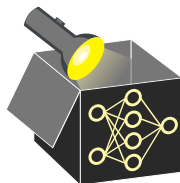
- **Intercept:** If all feature values are 0 (and season is WINTER $\hat{=}$ reference cat.), the expected number of bike rentals is $\hat{\theta}_0 = 3229.3$

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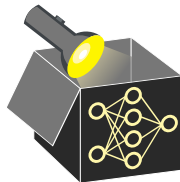
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- **Numerical:** Rentals increase by $\hat{\theta}_4 = 120.5$ if temp increases by 1 °C, c.p.